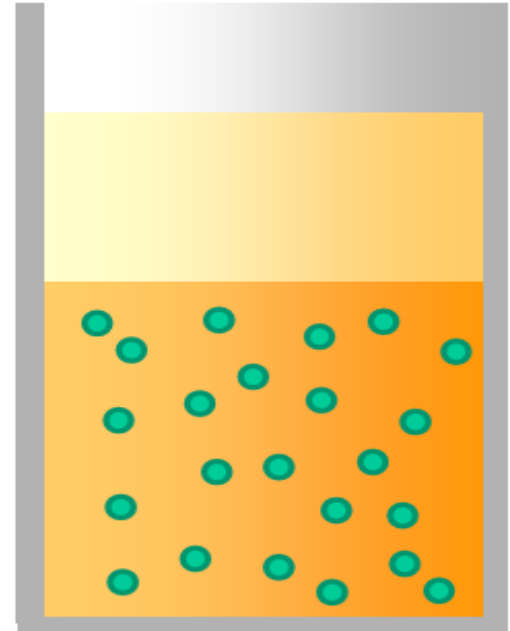


COLLOIDS

COLLOIDS

- A colloid is a substance microscopically dispersed throughout another substance.
- The word colloid comes from a Greek word '**kolla**', which means glue thus colloidal particles are glue like substances.
- These particles pass through a filter paper but not through a semipermeable membrane.
- Colloids can be made settle by the process of centrifugation.

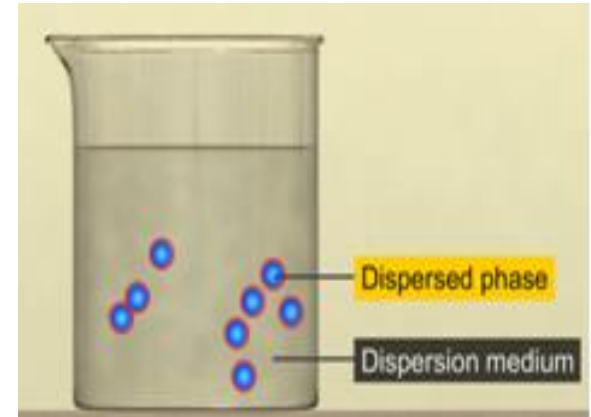


COLLOIDS...

➤ The colloidal system consist of two phases:

A dispersed phase (A discontinuous phase)

A dispersion medium (A continuous phase)

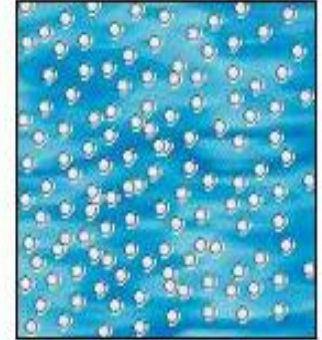


- The dispersed-phase particles have a diameter of between approximately **1nm – 100nm** .
- Such particles are normally invisible in an optical, though their presence can be confirmed with the use of an **ultramicroscope or an electron microscope**.

COLLOIDS...

SOLUTIONS

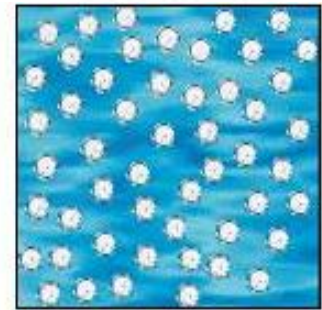
- ❑ Made up of particles or solutes and a solvent
- ❑ The solvent part of the solution is usually a liquid, but can be a gas.
- ❑ The particles are atoms, ions, or molecules that are **very small in diameter**.



True solution

COLLOIDAL MIXTURE

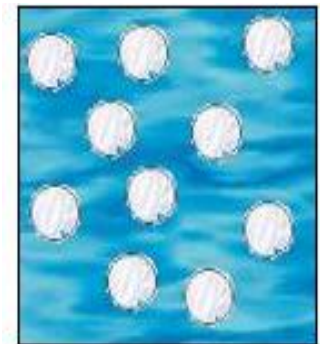
- ❑ Has particles that are not as small as a solution and not as large as a suspension.
- ❑ The particles **are intermediate in size**.



Colloidal solution

SUSPENSIONS

- ❑ Made up of particles and a solvent
its **particles are larger** than those found in a solution.
- ❑ The particles in a suspension can be distributed throughout the suspension evenly by shaking the mixture.



Suspensions



Solution:

Table salt dissolves in water to form Salt (saline) water. A solute (salt; NaCl) is dissolved in another substance (water) known as a solvent, and this creates a solution.



Suspension:

Flour suspended in water (appears light blue because blue light is scattered off the flour particles to a greater extent than red light)



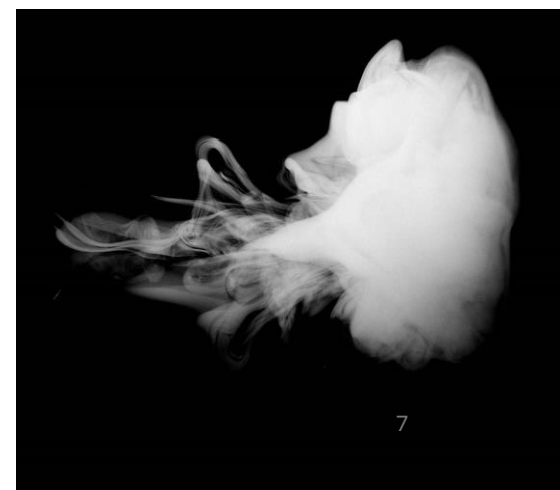
Colloid:

Milk is an emulsified colloid of liquid butterfat globules dispersed within a water based liquid. Colloids are Stabilized In suspension by Electrostatics - mutual Repulsion of like electrical Charges.

Comparison of the Properties of Solutions, Colloids, And Suspensions

Property	True Solution	Colloid	Suspension
Particle Size	Less than 1 nm	1 to 100 nm	More than 100 nm
Appearance	Clear	Cloudy	Cloudy
Homogeneity	Homogeneous	Homogeneous or Heterogeneous	Heterogeneous
Transparency	Transparent but often coloured	Often translucent and opaque but can be transparent	Often opaque but can be translucent
Separation	Does not separate	Can be separated	Separates or settles
Filterability	Passes through filter paper	Passes through filter paper	Particles do not pass through filter paper

- Examples of colloids are milk, synthetic polymers, fog, blood, jam, shoe polish, smoke, etc.



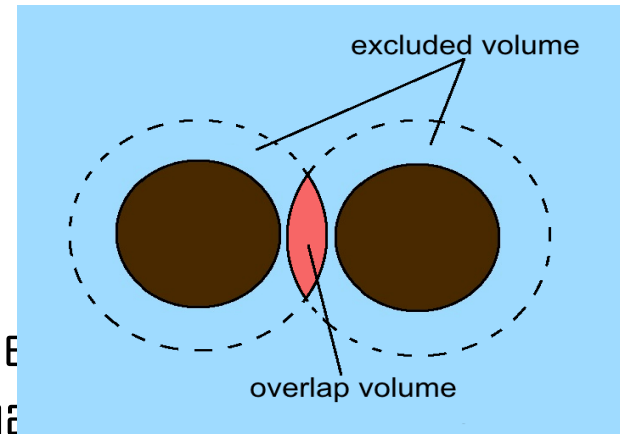
INTERACTION BETWEEN PARTICLES

The following forces play an important role in the interaction of colloid particles:

EXCLUDED VOLUME REPULSION IN LIQUID THEORY :

In liquid state theory, the 'excluded volume' of a molecule is the volume that is **inaccessible to other molecules** in the system as a result of the presence of the first molecule.

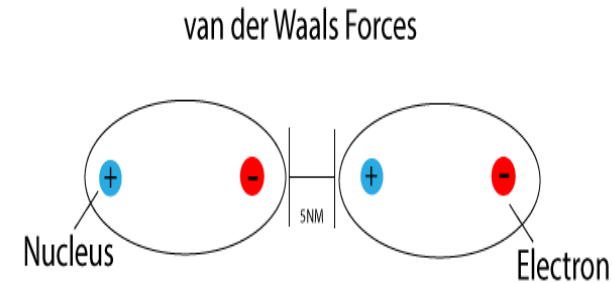
The excluded volume of a hard sphere is eight times its volume—however, for a two-molecule system, this volume is distributed among the two particles, giving conventional result of four times the volume.



INTERACTION BETWEEN PARTICLES

VAN DER WAALS FORCE :

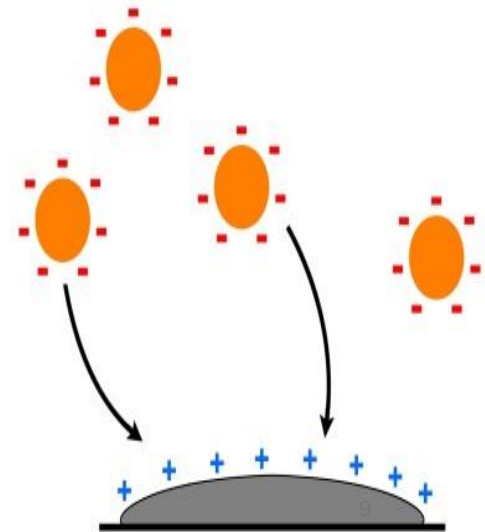
It is the sum of the attractive or repulsive **forces between molecules** other than those due to covalent bonds, the hydrogen bonds, or the electrostatic interaction of ions with one another or with neutral molecules or charged molecules.



ELECTROSTATIC INTERACTION :

Colloidal particles often carry an electrical charge and therefore **attract or repel each other**.

The charges of both the continuous and the dispersed phase, as well as the mobility of the phases are factors affecting this interaction.

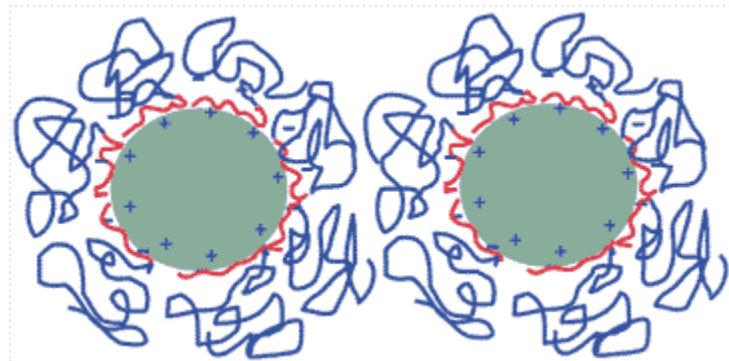


INTERACTION BETWEEN PARTICLES

STERIC FORCES :

Steric effects arise from the fact that each atom within a molecule occupies **a certain amount of space**.

If atoms are brought too close together, there is an associated cost in energy due to overlapping electron clouds (Pauli or Born repulsion), and this may affect the molecule's preferred shape (conformation) and reactivity.



CLASSIFICATION OF COLLOIDS

- ❑ Based of physical state of dispersed phase an dispersion medium.
- ❑ Based of nature of interaction between dispersed phase and dispersion medium.
- ❑ Based on molecular size in the dispersed phase.
- ❑ Based on appearance of colloids.
- ❑ Based on electric charge on dispersion phase.

Based on physical state of dispersed phase and dispersion medium

Dispersed Phase	Dispersion Medium	Name	Examples
Solid	Solid	Solid-Sol	Alloys, Cranberry glass
Solid	Liquid	Sol	Ink, Blood
Solid	Gas	Aerosol	Smoke, Ice cloud
Liquid	Solid	Gel	Jelly, Curd
Liquid	Liquid	Emulsion	Milk, Cream
Liquid	Gas	Liquid aerosol	Cloud, Fog
Gas	Solid	Solid form	Aerogel, Pumice stone
Gas	Liquid	Foam	Shaving cream
Gas	Gas	None	All gases are miscible

BASED ON NATURE OF INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM

LYOPHILIC COLLOIDS

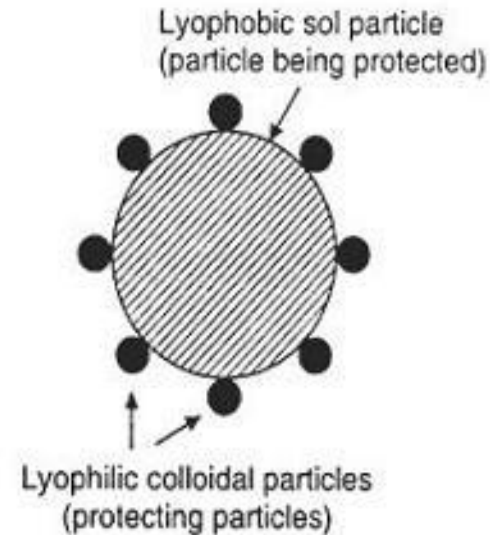
- Colloidal solution in which the dispersed phase has a great affinity for the dispersion medium.
- They are also termed as **intrinsic colloids**.
- Such substances have tendency to pass into colloidal solution when brought in contact with dispersion medium.
- If the dispersion medium is water, they are called **hydrophilic or emulsoids**.
- The lyophilic colloids are generally **self- stabilized**.
- **Reversible** in nature and are heavily hydrated.
- Example of lyophilic colloids are starch, gelatin, rubber, protein etc.



BASED ON NATURE OF INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM

LYOPHOBIC COLLOIDS

- Colloidal solutions in which the dispersed phase has no affinity to the dispersion medium.
- These are also referred as **extrinsic colloids**.
- Such substances have no tendency to pass into colloidal solution when brought in contact with dispersion medium.
- The lyophobic colloids are relatively **unstable**.
- They are **irreversible** by nature and are stabilized by adding small amount of electrolyte.
- They are poorly hydrated.
- If the dispersion medium is water, the lyophobic colloids are called **hydrophobic or suspensoids**.
- Examples: sols of metals like Au, Ag, sols of metal hydroxides and sols of metal sulphides.



BASED ON NATURE OF INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM

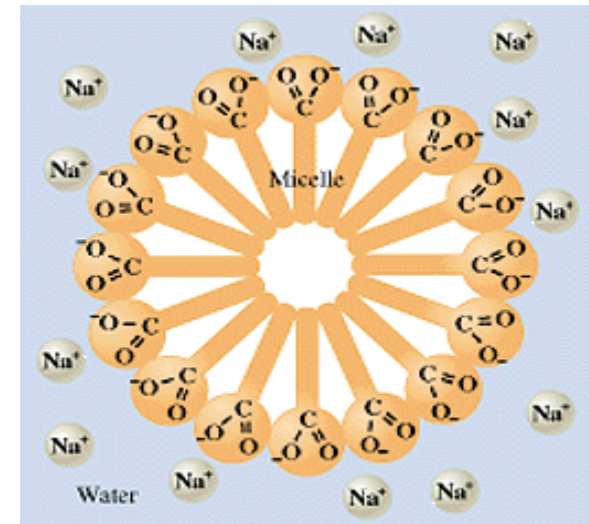
ASSOCIATED COLLOIDS

These colloids behave as normal electrolytes at low concentrations but behave as colloids at higher concentrations.

These associated colloids are also referred to as **micelles**.

Sodium stearate ($\text{C}_{18}\text{H}_{35}\text{NaO}_2$) behave as electrolyte in dilute solution but colloid in higher concentrations.

Examples: Soaps , higher alkyl sulphonates , polythene oxide.



Based on molecular size in the dispersed phase.

➤ MULTIMOLECULAR COLLOIDS

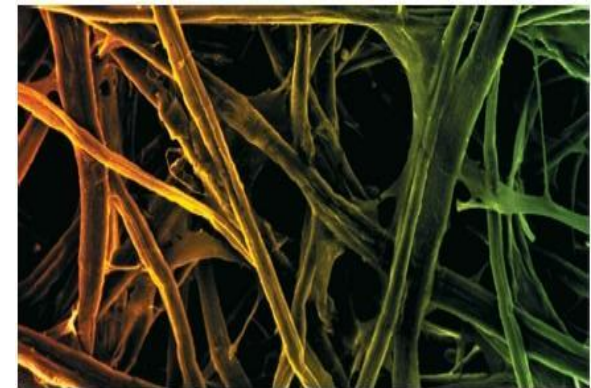
Individual particles of the dispersed phase consists of **aggregates of atoms or small molecules** having diameter less than 10^{-7}cm . The particles are held by weak **vander waal's forces**.

Example; gold sol, sulphur sol

➤ MACROMOLECULAR COLLOIDS

The particles of dispersed phase are sufficiently **large in size** enough to be of colloidal solution. These are called Natural Polymers.

Examples are starch, cellulose and proteins.



Based on appearance of colloids

• SOLS

- When a colloidal solution appears as **fluid**.
- The sols are generally named as dispersion medium.
- When the dispersion medium is water, the sol is known as **hydrosol or aquosol**.
- When the dispersion medium is alcohol or benzene it is called **alcosol and benzosol** respectively.



• GELS

- When a colloidal solution appears as **solid**.
- The rigidity of gel varies from substance to substance.
- Examples : jelly, butter, cheese, curd.

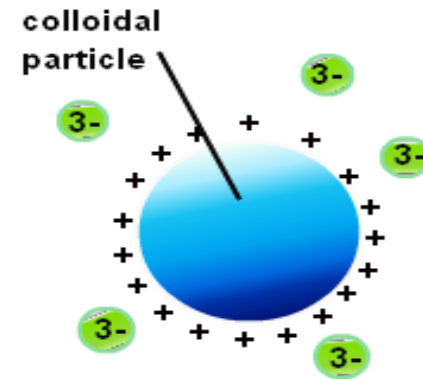


BASED ON ELECTRICAL CHARGE ON DISPERSION PHASE

➤ POSITIVE COLLOIDS

When dispersed phase in a colloidal solution carries **a positive charge**.

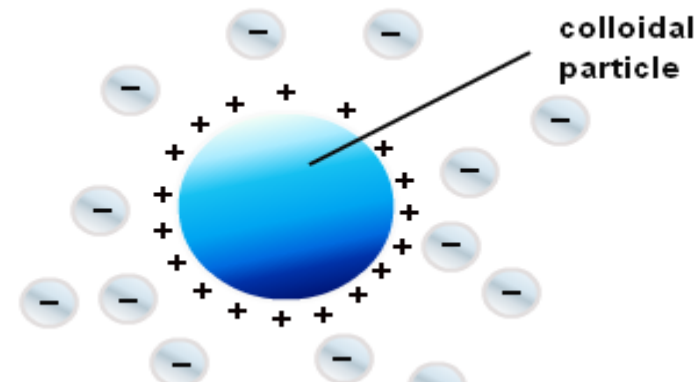
Examples : Metal hydroxides like $\text{Fe}(\text{OH})_3$, $\text{Al}(\text{OH})_3$, methylene blue sol etc.



➤ NEGATIVE COLLOIDS

When dispersed phase in a colloidal solution carries **a negative charge**.

Examples : Ag sol, Cu sol



PREPARATION OF COLLOIDS

❖ *Mechanical Dispersion*

❖ *Electrical dispersion or Bredig's Arc Method*

❖ *Peptisation*

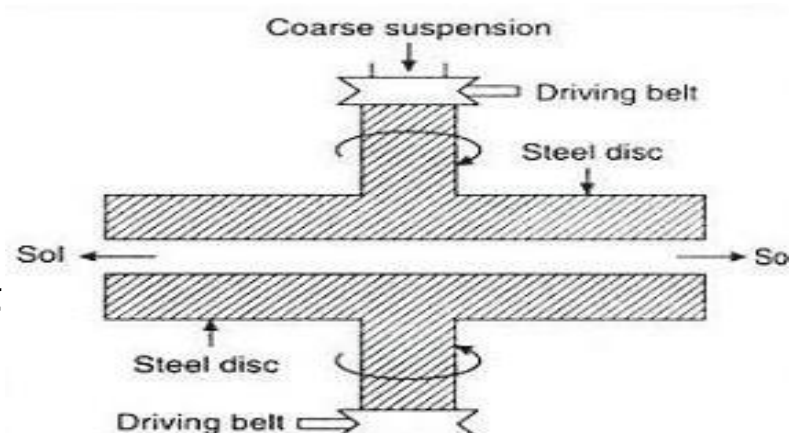
❖ *Condensation Method*

PREPARATION OF COLLOIDS...

(A) Mechanical dispersion:

In this method,

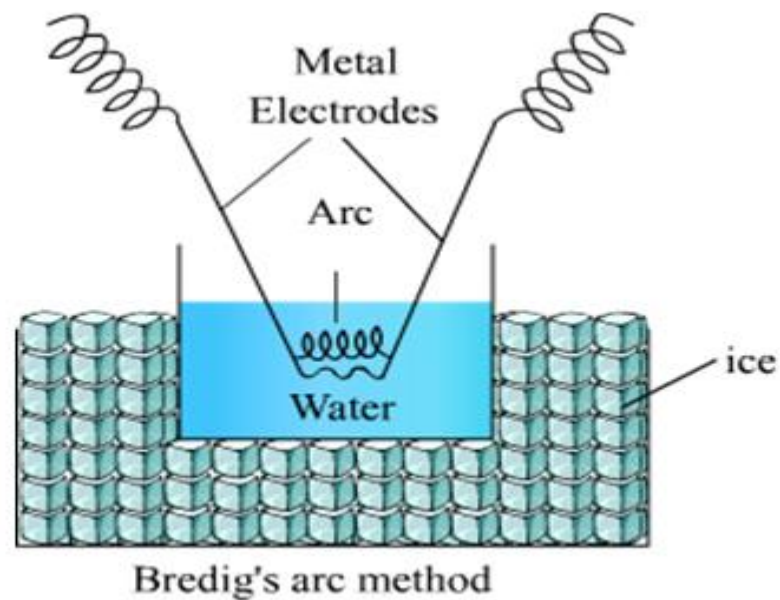
- The substance is first ground to coarse particles.
- It is then mixed with the dispersion medium to get a suspension.
- The suspension is then grinded in colloidal mill.
- It consists of two metallic discs nearly touching each other and rotating in opposite directions at a very high speed about 7000 revolution per minute.
- The space between the discs of the mill is so adjusted that coarse suspension is subjected to great shearing force giving rise to particles of colloidal size.
- Colloidal solutions of black ink, paints, varnishes, dyes etc. are obtained by this method.



PREPARATION OF COLLOIDS...

(B) By electrical dispersion or Bredig's arc method:

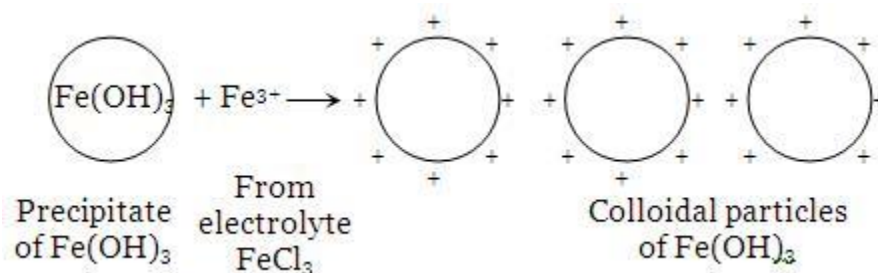
- This method is used to prepare sols of platinum, silver, copper or gold.
- The metal whose sol is to be prepared is made as two electrodes which immerse in dispersion medium such as water etc.
- The dispersion medium is kept cooled by ice.
- An electric arc is struck between the electrodes.
- The tremendous heat generated by this method give colloidal solution.
- The colloidal solution prepared is stabilized by adding a small amount of KOH to it.



PREPARATION OF COLLOIDS...

(C) By peptisation:

- The process of converting a freshly prepared precipitate into colloidal form by **the addition of suitable electrolyte** is called peptisation.
- Cause of peptisation is the adsorption of the ions of the electrolyte by the particles of the precipitate.
- The electrolyte used for this purpose is called peptizing agent or stabilizing agent.
- Important peptizing agents are **sugar, gum, gelatin and electrolytes**.



PREPARATION OF COLLOIDS...

(D) Condensation method

- In condensation method, the smaller particles of the dispersed phase are aggregated to form larger particles of colloidal dimensions.
- *Some important condensation methods are described below:*
 - a) Solutions of substances like mercury and sulphur are prepared by passing their vapours through a cold water containing a suitable stabilizer such as ammonium salt or citrate.



PREPARATION OF COLLOIDS...

(b) By excessive cooling:

A colloidal solution of ice in an organic solvent like ether or chloroform can be prepared by freezing a solution of water in solvent.

The molecules of water which can no longer be held in solution,
separately combine to form particles of colloidal size.

c) By exchange of solvent:

Colloidal solution of certain substances such as sulphur, phosphorus which are soluble

in alcohol but insoluble in water can be prepared by pouring their alcoholic solution in
excess of water.

For example alcoholic solution of sulphur on pouring into water gives milky colloidal
solution of sulphur.

PREPARATION OF COLLOIDS...

d) Chemical methods:

Colloids can be prepared by following chemicals methods.

1) Oxidation:

Addition of oxygen and removal of hydrogen is called oxidation.

For example: Colloidal solution of sulphur can be prepared by oxidizing an aqueous solution of H₂S with a suitable oxidizing agent such as bromine water.



PREPARATION OF COLLOIDS...

2) Reduction:

Addition of hydrogen and removal of oxygen is called reduction.

For example: Gold sol can be obtained by reducing a dilute aqueous solution of gold with stannous chloride.



3) Hydrolysis:

It is the break down of water.

Sols of ferric hydroxide and aluminium hydroxide can be prepared by boiling the aqueous solution of the corresponding chlorides.

For example.



PREPARATION OF COLLOIDS...

4) Double Decomposition

The sols of inorganic insoluble salts such as arsenous sulphide, silver halide etc may be prepared by using double decomposition reaction.

For example: Arsenous sulphide sol can be prepared by passing H₂S gas through a dilute aqueous solution of arsenous oxide.



Purification of colloids

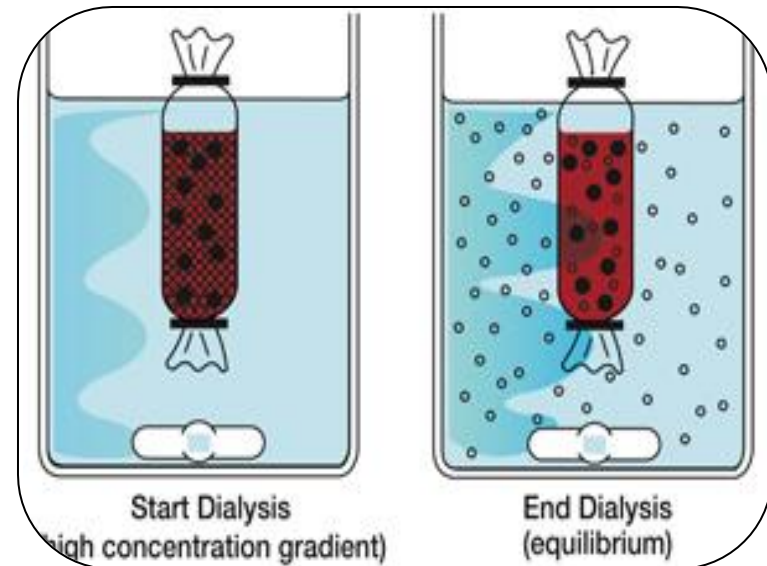
There are three common methods used for purification of colloids:

- Dialysis
- Electrodialysis
- Ultra filtration

Purification of colloids...

Dialysis

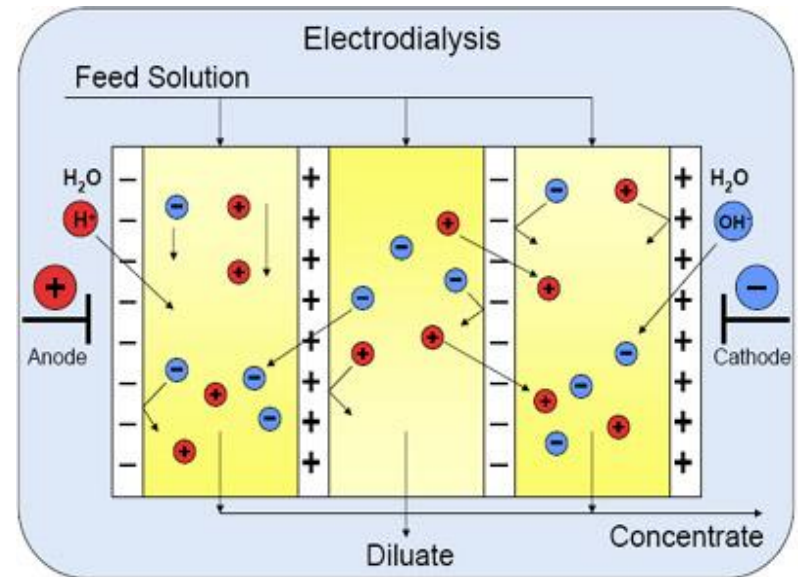
- The process of separating the particles of colloid from those of crystalloid, by means of diffusion through a suitable membrane.
- Its principle is that **colloidal particles can not pass through a cellophane membrane while the ions of the electrolyte can pass through it.**
- The impurities slowly diffused out of the bag leaving behind pure colloidal solution.
- The distilled water is changed frequently to avoid accumulation of the crystalloids otherwise they may start diffusing back into the bag.
- Dialysis can be used for removing HCl from the ferric hydroxide sol.



Purification of colloids...

Electro dialysis

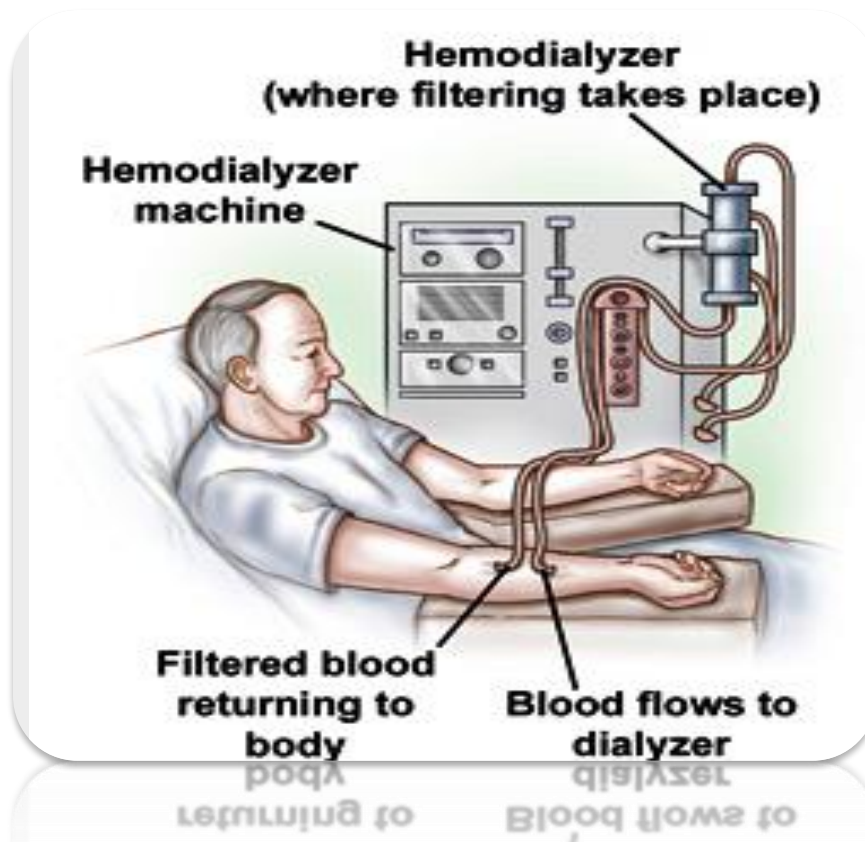
- The process of dialysis is very slow.
- The process is speeded up by application of electrical potential.
- This is called electro dialysis.



Purification of colloids...

APPLICATION OF ELECTRO DIALYSIS:

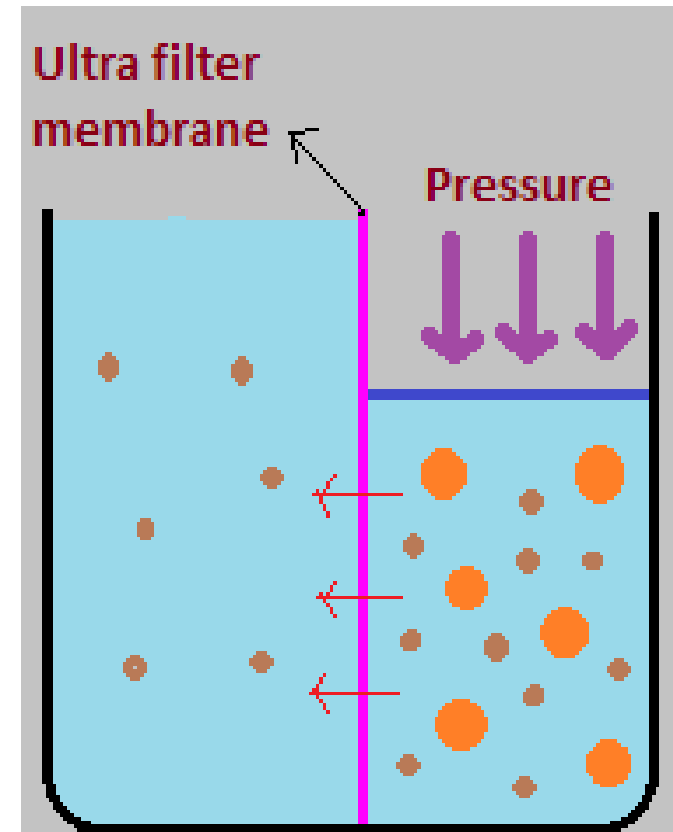
Artificial kidney machine make use of electro dialysis.



Purification of colloids...

Ultra filtration

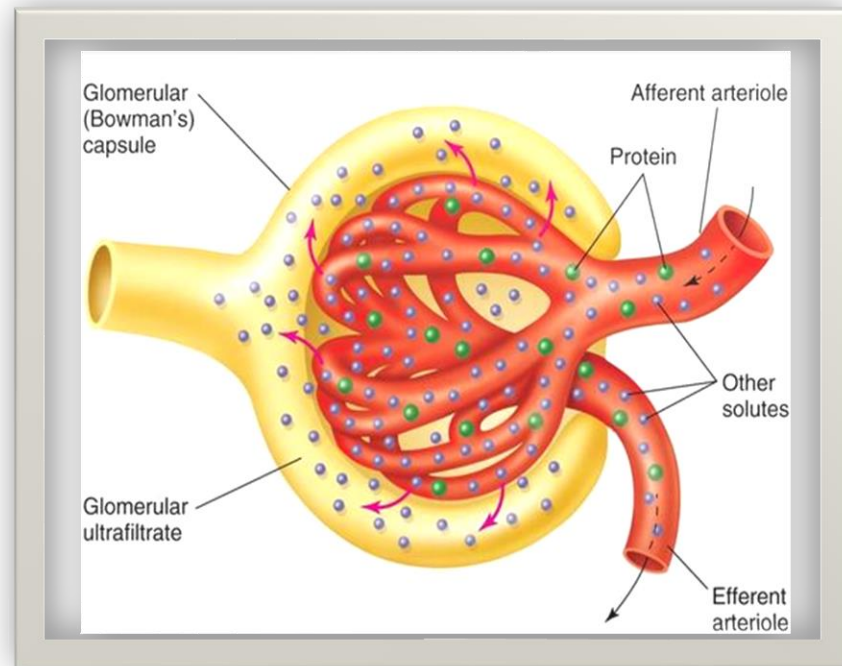
- Ultra filtration is a process of high pressure filtration through a semi permeable membrane in which colloidal particles are retained while the small sized solutes and the solvent are forced to move across the membrane by hydrostatic pressure forces.
- The process makes use of the filter paper made up of nitrocellulose or colloidons.



Purification of colloids...

Application of ultra filtration:

Ultra filtration is a vital process that takes place in the kidneys.



PROPERTIES OF COLLOIDS

PHYSICAL PROPERTIES OF COLLOIDS

- **Heterogeneity:** Colloidal solutions consist of two phases-dispersed phase and dispersion medium.
- **Visibility of dispersed particles:** The dispersed particles present in them are not visible to the naked eye and they appear homogenous.
- **Filterability:** The colloidal particles pass through an ordinary filter paper. However, they can be retained by animal membranes, cellophane membrane and ultrafilters.
- **Stability:** Lyophilic sols in general and lyophobic sols in the absence of substantial concentrations of electrolytes are quite stable.
- **Colour:** The colour of a colloidal solution depends upon the size of colloidal particles present in it. Larger particles absorb the light of longer wavelength and therefore transmit light of shorter wavelength.

OPTICAL PROPERTIES OF COLLOIDS

• TYNDALL EFFECT

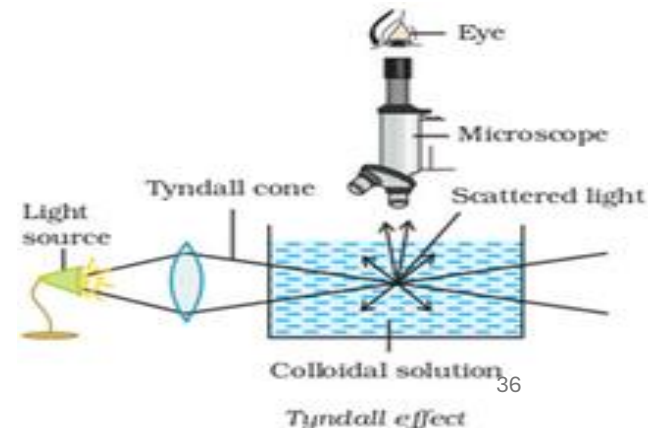
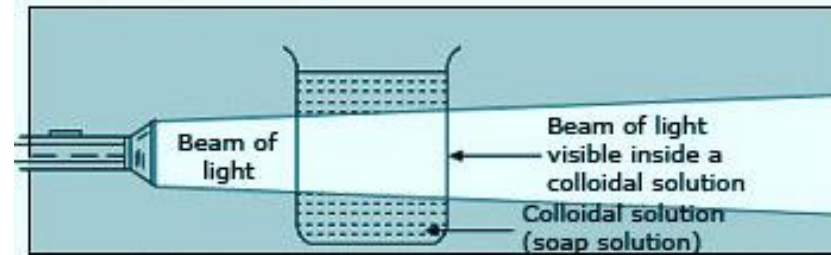
- When an intense converging beam of light is passed through a colloidal solution kept in dark, the path of the beam gets illuminated with a bluish light.

- This phenomenon is called Tyndall effect and the illuminated path is known as **Tyndall cone**.

- The Tyndall effect is due to the **scattering of light by colloidal particles**.

- Tyndall effect is not exhibited by true solutions. This is because the particles present in a true solution are too small to scatter light.

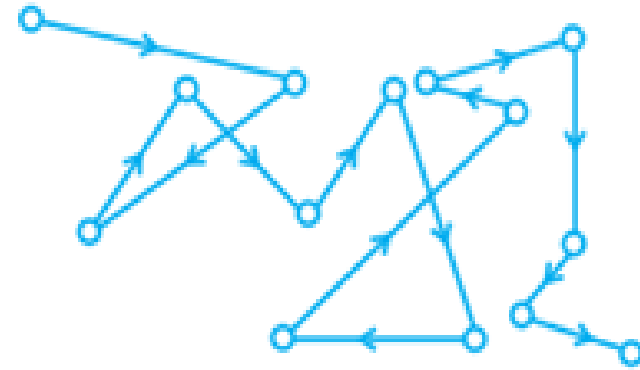
- Tyndall effect can be used to distinguish a colloidal solution from a true solution. The phenomenon has also been used to devise an instrument known as ultra microscope. The instrument is used for the detection of the particles of colloidal dimensions.



MECHANICAL PROPERTIES OF COLLOIDS

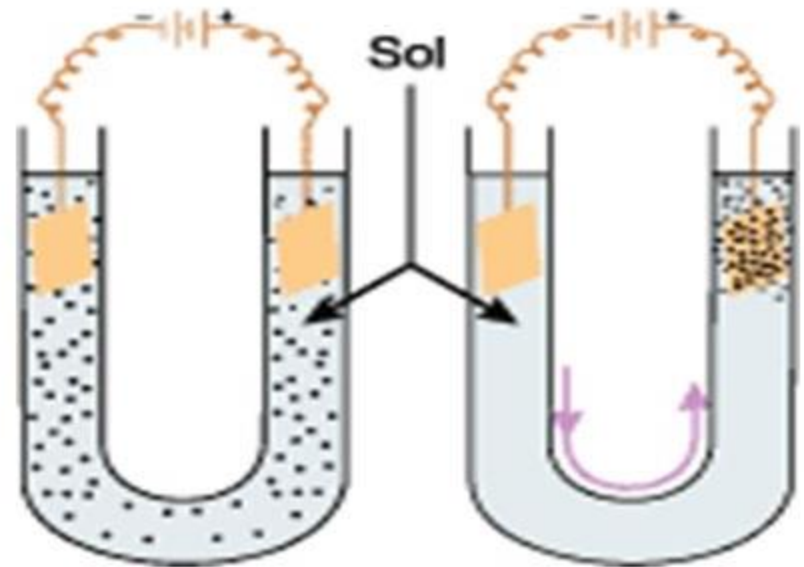
• BROWNIAN MOVEMENT

- The **continuous zigzag movement** of the colloidal particles in the dispersion medium in a colloidal solution is called Brownian movement.
- Brownian movement is due to the unequal bombardments of the moving molecules of dispersion medium on colloidal particles.
- The Brownian movement decreases with an increase in the size of colloidal particle. This is why suspensions do not exhibit this type of movement.



ELECTRICAL PROPERTIES OF COLLOIDS

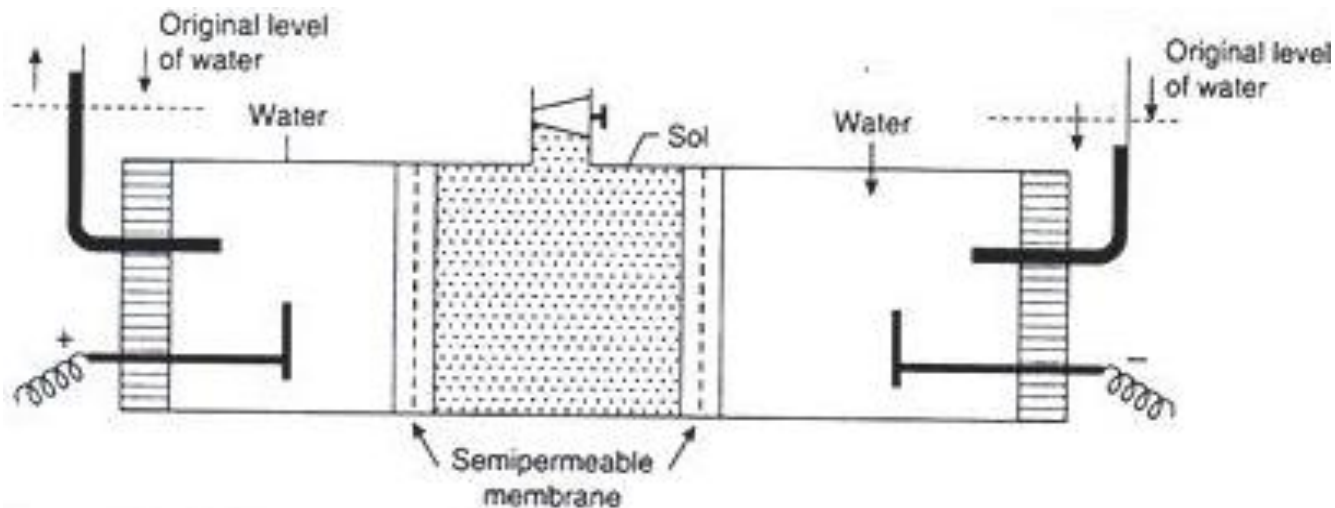
- **ELECTROPHORESIS**
- The movement of colloidal particles towards a particular electrode under the influence of an electric field.
- If the colloidal particles carry positive charge, they move towards cathode when subjected to an electric field and vice versa.



ELECTRICAL PROPERTIES OF COLLOIDS

- **ELECTROSMOSIS**

- The movement of dispersion medium under the influence of an electric field in the situation when the movement of colloidal particles is prevented with the help of a suitable membrane.
- During electrosmosis, colloidal particles are checked and it is the dispersion medium that moves towards the oppositely charged electrode.

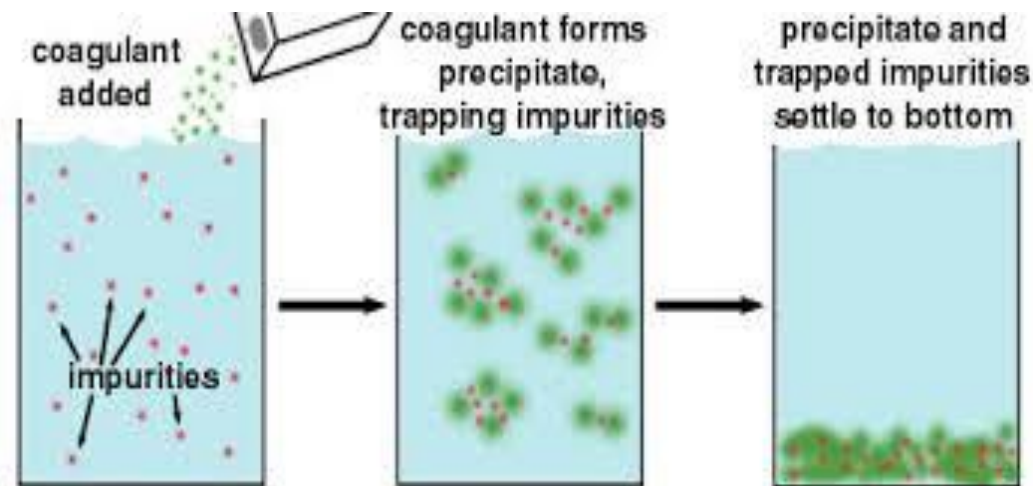


• COAGULATION OR FLOCCULATION

- Coagulation may be defined as the phenomenon involving the precipitation of a colloidal solution on addition of an electrolyte.

- **Hardy-Schulze rule**

The greater is the valence of the oppositely charged ion of the electrolyte added to a colloidal solution, the faster is the coagulation of the colloidal solution.



- **Flocculation value:** The coagulating power of an electrolyte is usually expressed in terms of its flocculation value which may be defined as the minimum concentration (in millimoles per litre) of an electrolyte required to cause the coagulation of a sol.
- A smaller flocculation value indicates the greater coagulating power of the electrolyte. Thus,

$$\text{Coagulating power} \propto \frac{1}{\text{Flocculation value}}$$

- The coagulation of colloidal solution can also be achieved by any of the following methods.

By electrophoresis

By mixing two oppositely sols

By persistent dialysis

ADVANTAGES OF COLLOIDS

- Colloids allow the dispersion of normally insoluble materials, such as metallic gold or fats. These can then be used more easily, or absorbed more easily.
- Colloidal gold, for example, can be used in medicine to carry drugs and antibiotics, because it is highly non-reactive and non-toxic.
- Pharmaceutical industry makes use of colloidal solution preparation in many medicines. A wide variety of medicines are emulsions. An example is Cod Liver Oil.

